

DESIGN

EXECUTION



Vision meets **structural** execution

NEXO Engineering & Contracting designs, builds, and teaches — architectural and structural modelling, heavy civil contracting, and the technical training that keeps both standing.

A-01

Architectural & structural design

C-02

General contracting

T-03

Training academy



What this document sets out

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Next-Generation Engineering eXcellence Organization

Rooted in Baghdad, NEXO closes the gap between the model and the site.

We bring BIM-grade technical modelling, advanced construction chemistry, and local workforce development under one practice — so what gets drawn is what gets built.

Design and execution are not handed between companies here. They are two axes of the same office, meeting at a single engineered joint: the drawing that a crew can actually build, and the crew that can read it.



FIG. 01 — ON SITE, BAGHDAD

VISION

To be Iraq's leading multidisciplinary engineering practice, setting the benchmark for architectural modelling, structural durability, and technical education.

MISSION

Deliver turnkey civil contracting, architectural precision, and practical software training by pairing international engineering standards with strategic industrial partnerships.

WHERE WE WORK

Commercial, hospitality, petroleum, and urban development projects across Iraq — from concept package to handover.

HOW WE WORK

International codes and specifications, materials from established construction-chemistry manufacturers, and site supervision by the same engineers who produced the model.

THE NAME, AND THE MARK

Next-Generation Engineering eXcellence Organization — two heavy members crossing without merging, separated at the centre by a fine diagonal void. That void is the joint where design meets execution.

Three divisions, one practice

Each runs its own discipline and its own crew. They are parallel, not sequential — a client may engage any one of them alone, or all three across a single project life-cycle.

A-01	Architectural & Structural Design	05
	Concept packages and detailed calculations to international code.	

C-02	General Contracting & Construction	06
	Turnkey execution for commercial, petroleum, hospitality, and urban work.	

T-03	Training & Capacity Academy	07
	Industry-standard software and site expertise for engineers across Iraq.	



A-01

Architectural & Structural Design

Concept packages and detailed calculations to international code — delivered as a coordinated model, not a set of disconnected drawings.

SCOPE OF WORK

- ❑ Architectural spatial planning & 3D visualisation
- ❑ Structural calculations & BIM modelling
- ❑ Civil layout optimisation
- ❑ Exterior facade & interior engineering

DELIVERABLES

- ❑ Concept and schematic design packages
- ❑ Coordinated BIM model & clash review
- ❑ Structural calculation reports
- ❑ Tender-ready drawing sets & quantities



FIG. 02 — SETTING-OUT TRANSFERRED FROM MODEL TO SLAB; STRUCTURAL GRID AND PILE POSITIONS MARKED DIRECTLY ON THE RAFT PROTECTION LAYER



C-02

General Contracting & Construction

Turnkey execution for commercial, petroleum, hospitality, and urban work — from excavation and substructure through to handover.

SCOPE OF WORK

- High-rise structural construction
- Petroleum & industrial infrastructure
- MEP systems integration
- Structural rehabilitation & retrofitting

ON SITE

- Excavation, shoring & substructure works
- Waterproofing & protection systems
- Reinforcement, formwork & concrete placement
- Quality control, testing & HSE supervision

SITE CAPABILITIES DOCUMENTED IN THIS PROFILE

C-02.1

Backfilling works

P. 09

C-02.2

Raft waterproofing

P. 10

C-02.3

Blockwork

P. 12

C-02.4

Concrete coring

P. 13

C-02.5

Screed works

P. 14

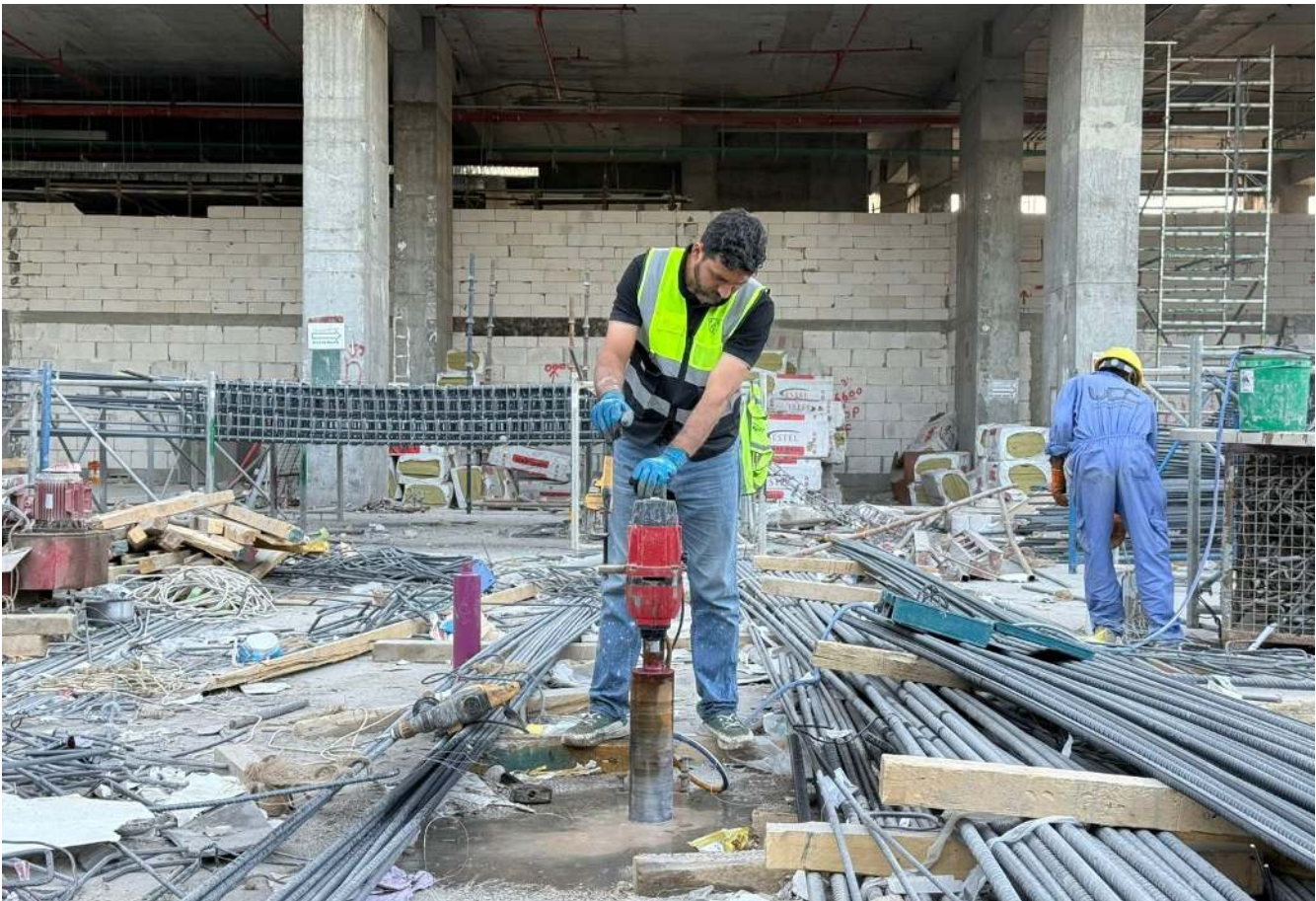


FIG. 03 – EXECUTION ACROSS TRADES: A PENETRATION CUT TO THE SERVICES DRAWINGS ON THE WORKING FLOOR, ALONGSIDE MATERIAL STAGED FOR THE TRADES THAT FOLLOW



T-03

Training & Capacity Academy

Industry-standard software and site expertise for engineers across Iraq — taught by the engineers running our sites, on the software the industry actually uses.

SCOPE OF WORK

- ❑ BIM — Revit, ETABS, AutoCAD, Primavera P6
- ❑ Site supervision & HSE standards
- ❑ Concrete testing & chemical application
- ❑ Technical masterclasses & certification

FORMAT

- ❑ Workshops for practising engineers & graduates
- ❑ In-house programmes for contractor teams
- ❑ Applied sessions on live project material
- ❑ Certificate of completion on assessment

LEARN IT WHERE IT GETS BUILT

The academy exists because the gap between a model and a buildable model is taught nowhere else in the market — so we teach it on our own drawings, on our own sites.



FIG. 04 — DETAIL TAUGHT ON SITE: PILE HEAD CUT DOWN TO CUT-OFF LEVEL, MEMBRANE DRESSED AND SEALED AT THE COLLAR, READY FOR NON-SHRINK GROUT

Built, not rendered

Five site capabilities — substructure waterproofing, earthworks, masonry, controlled drilling, and floor screed — executed by NEXO's own crews under division C—02. Each sheet sets out the scope, the method, and the failure the method is there to prevent.

C-02.1

Backfilling works

Earthworks — backfilling & compaction

09

C-02.2

Raft waterproofing & pile heads

Substructure — membrane joints & pile head treatment

10

C-02.3

Blockwork

Masonry — blockwork & partitions

12

C-02.4

Concrete coring works

Diamond core drilling — MS penetrations

13

C-02.5

Screed works

Floor finishes — cementitious screed

14

Backfilling works

DIVISION	PACKAGE	LOCATION	CONTROL
C—02 · General contracting	Earthworks — backfilling & compaction	Baghdad, Iraq	Layer thickness, moisture, compaction

Backfilling controls what the structure sits on. The scope runs from formation preparation and levels control through to material placed in specified lifts, moisture-conditioned and mechanically compacted, then verified against the project specification before any following trade takes possession of the surface.

SCOPE OF WORKS

- ☒ Preparation and inspection of the formation
- ☒ Setting-out and levels control
- ☐ Supply of approved backfill material
- ☐ Delivery and tipping onto the formation
- ☐ Spreading and trimming to a working layer
- ☒ Placement in defined layers to the specified lift
- ☒ Moisture conditioning ahead of compaction
- ☒ Mechanical compaction of each layer
- ☒ Verification of levels and compaction
- ☒ Handover of a surface built to specification

PREVENTS Differential settlement under load — the defect that surfaces months later, once slabs and structures are in service and the fill beneath them can no longer be reached.



FIGS. 05-08 — MATERIAL PLACED AND SPREAD ACROSS THE PREPARED FORMATION; LEVELS CHECKED AT DELIVERY, AT TIPPING, AND AT SETTING-OUT

Raft waterproofing & pile head treatment

DIVISION	PACKAGE	LOCATION	SYSTEMS
C-02 · General contracting	Substructure — membrane, joints & pile heads	Baghdad, Iraq	Fosroc HDPE membrane · non-shrink grout

Raft waterproofing in Fosroc HDPE membrane, carried across the full footprint, lapped and sealed, and dressed into every construction joint and upstand. Each pile head is treated as an individual detail — cut down to cut-off level, reinforcement exposed and cleaned, the membrane sealed at the collar, then grouted, finished flush and cured to the approved method statement.

SCOPE OF WORKS

- ☑ Surface preparation of the blinding
- ☑ Treatment of construction joints and upstands
- ☐ Membrane laid, lapped and sealed to full footprint
- ☐ Membrane dressed into trenches and penetrations
- ☐ Protection layer before reinforcement on the deck
- ☑ Pile head cut-down to cut-off level
- ☑ Reinforcement exposed and cleaned
- ☑ Membrane termination and sealing at each collar
- ☑ Formwork to pile head collars where required
- ☑ Non-shrink grout placed, finished and cured

PREVENTS A breach at a pile head — where a tanked raft normally fails, and unreachable once the raft is poured and the superstructure is up. This is the last point at which the envelope can be inspected before it disappears under concrete.



FIGS. 09-11 — MEMBRANE LAID, LAPPED AND SEALED TO THE FULL FOOTPRINT; DRESSED INTO EVERY CHANGE OF PLANE; EACH PILE HEAD DETAILED INDIVIDUALLY — PLATE OVERLEAF

On site

BAGHDAD, IRAQ
DIVISION C-02



FIGS. 12-13 – REINFORCEMENT PROCEEDING ABOVE AN INTACT WATERPROOFING ENVELOPE; SETTING OUT CARRIED OVER THE PROTECTED SURFACE, EVERY PILE POSITION COLLARED

Blockwork

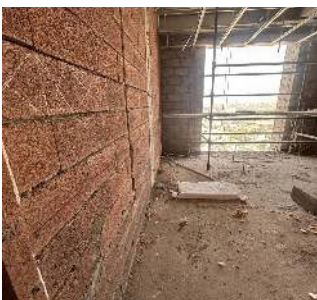
DIVISION	PACKAGE	LOCATION	INTERFACES
C—02 · General contracting	Masonry — blockwork & partitions	Baghdad, Iraq	Structural frame, M&E first fix, finishes

Blockwork is the reference layer every following trade sets out from. Block is coursed to line and level in the specified bond, the bay closed to the underside of the slab, the junction with the structural frame made good, and chase positions marked with the M&E installation before the wall is handed over for finishes.

SCOPE OF WORKS

- ☑ Setting out wall lines and openings from the grid
- ☑ Mortar or adhesive to the bedding face
- ☐ Laying block to line and level in the specified bond
- ☐ Cutting and fitting at returns and reveals
- ☐ Coursing to the underside of slab or beam
- ☑ Forming openings for doors, windows and services
- ☑ Making good at columns and structural elements
- ☑ Marking chase positions for M&E services
- ☑ Fitting accessories to complete the wall
- ☑ Handover of a wall ready for finishes

PREVENTS A wall out of plumb or off-line — inherited by plaster thickness, frame fit and service routing long after the mason has moved on; and cracking at the frame junction where movement was not allowed for.



FIGS. 14-18 — BONDED CORNER; SETTING OUT ON THE BLOCK FACE; THE BAY CLOSED TO FULL HEIGHT; THE JUNCTION WITH THE FRAME MADE GOOD; CHASES MARKED FOR SERVICES

Concrete coring works

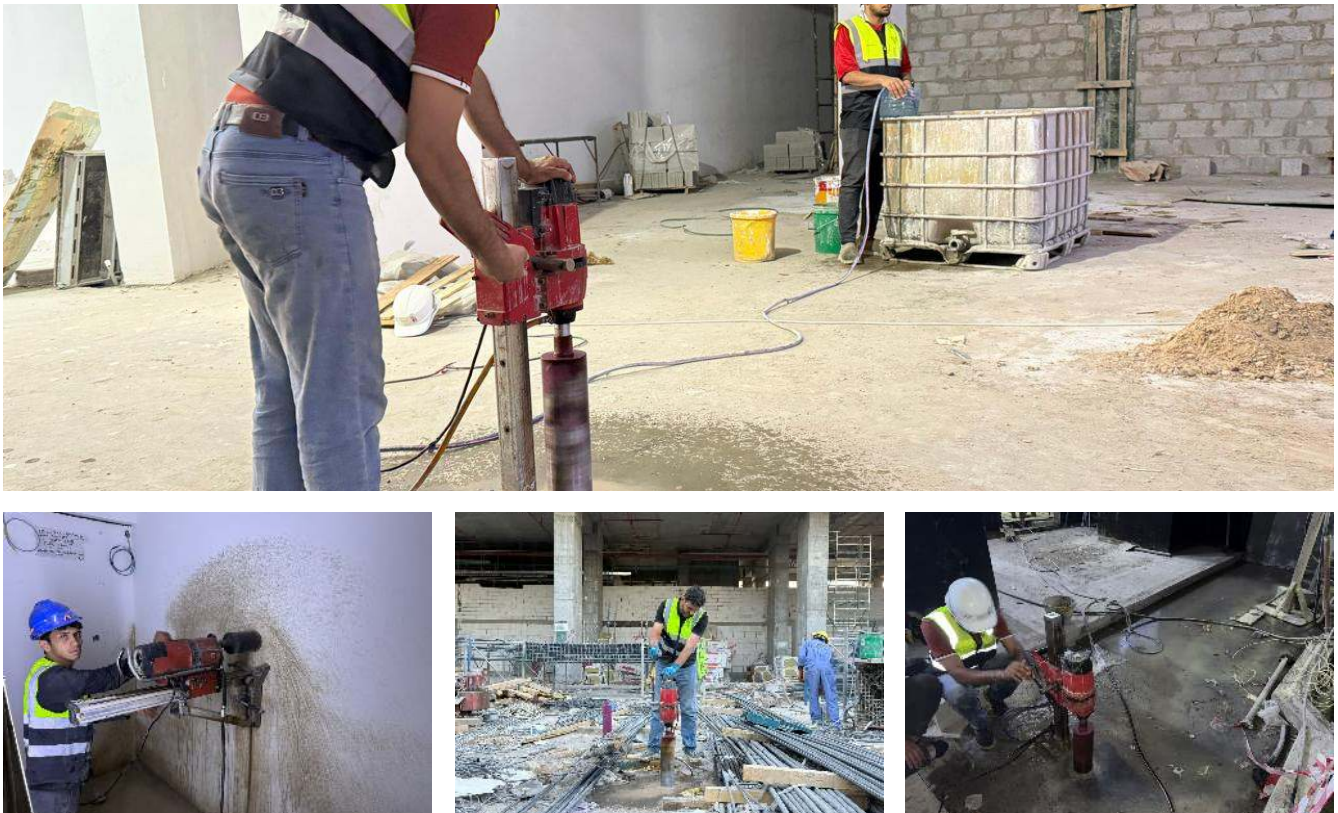
DIVISION	PACKAGE	LOCATION	METHOD
C—02 · General contracting	Diamond core drilling — MEP penetrations	Baghdad, Iraq	Rig fixed to the structure · water-fed

Diamond core drilling opens a controlled, dimensioned path through a reinforced element for MEP services. The penetration is set out against the drawings and confirmed clear of primary reinforcement, the rig fixed to the structure, and the cut made under controlled feed with water fed continuously through the bit.

SCOPE OF WORKS

- ☑ Setting out the penetration against the MEP drawings
- ☑ Confirming the position clear of primary reinforcement
- ☐ Fixing the rig — floor stand or wall-mounted frame
- ☐ Connecting the water feed for cooling and flushing
- ☐ Drilling under controlled feed, vertical or horizontal
- ☑ Managing water-fed slurry at the point of cut
- ☑ Removal of the drilled core from the barrel
- ☑ Cleaning arisings from the finished opening
- ☑ Protection of surrounding structure and finishes
- ☑ Handover of a clean, dimensioned penetration

PREVENTS A severed bar, or a spalled and slurry-flooded opening — permanent damage, since nothing cut out of a reinforced element can be put back.



FIGS. 19-22 — RIG FIXED TO THE STRUCTURE OVER A DIMENSIONED PENETRATION; WATER FED CONTINUOUSLY THROUGH THE BIT

Screed works

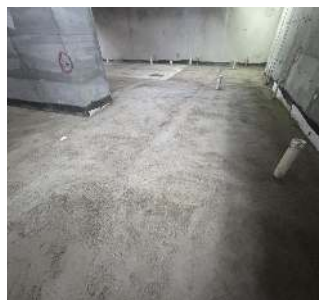
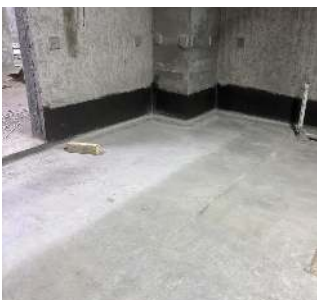
DIVISION	PACKAGE	LOCATION	CONTROL
C—02 · General contracting	Floor finishes — cementitious screed	Baghdad, Iraq	Datum points, set joints, specified tolerance

Screed converts a structural slab into the flat, level base the specified floor finish is laid on. Datum points are fixed before placing, levels and step transitions struck to tolerance, joint positions set rather than cut in afterwards, and the surface cured before it is handed over to the finishing trade.

SCOPE OF WORKS

- | | |
|---|--|
| ☒ Inspection and preparation of the substrate | ☒ Levelling and compacting to the set datum |
| ☒ Datum points and levels set across the bay | ☒ Finishing to the specified tolerance and texture |
| ☐ Slopes and step transitions set out | ☒ Formation of falls where the design requires |
| ☐ Mixing to the specified mix | ☒ Construction and control joints at set positions |
| ☐ Placing across the prepared substrate | ☒ Curing, and preparation to receive the finish |

PREVENTS An out-of-tolerance floor inherited by the tile, vinyl or resin laid on it — correctable only by taking the floor up again — and shrinkage cracking wherever an unjointed bay is restrained.



FIGS. 23–27 — THE FORMED STEP HELD TO THE SAME TOLERANCE AS THE OPEN FLOOR; ONE CONTINUOUS PLANE WITH A CONTROLLED JOINT; DATUM PINS SET BEFORE PLACING; THE WORKED SURFACE; THE PLANE CARRIED THROUGH THE CORRIDORS

Four trades, one crew structure

BAGHDAD, IRAQ
DIVISION C-02



FIGS. 28-31 — FILL PLACED AND COMPACTED TO FORMATION, BLOCK COURSED TO THE SLAB, PENETRATIONS CUT TO THE SERVICES DRAWINGS, AND FLOORS SCREEDED FLAT TO DATUM

Who we build with

Developers, energy operators, and the construction-chemistry suppliers behind the work.

CLIENTS

 Rixos Baghdad Luxury hospitality & residences CLIENT	 EBS Petroleum Energy & industrial CLIENT	 Millennium Tower Ramlah urban developments CLIENT	 Future City Modon developments CLIENT
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PARTNERS & SUPPLIERS

 Sika Construction chemicals SUPPLIER	 Fosroc Constructive solutions · Saint-Gobain SUPPLIER	 DCP Don Construction Products SUPPLIER	 Zamzam Land Development partner PARTNER
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SECTORS SERVED

S-01 Commercial Offices, retail, and mixed-use developments	S-02 Hospitality Hotels and serviced residences	S-03 Petroleum & industrial Energy infrastructure and plant facilities	S-04 Urban development Master-planned districts and towers
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Supplier relationships are how specification survives contact with the site. Materials are drawn from established construction-chemistry manufacturers and applied by crews trained on those systems.

Start a project with us

Tender documents, an engineering question, or a place on the next workshop — reach the Baghdad office directly.

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